

claude-windows / 8d4866bf-8edd-4da9-909e-022ddee12675

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\8d4866bf-8edd-4da9-909e-022ddee12675\subagents\workflows\wf_e27060ad-65f\agent-a4dade83fd3303835.jsonl`  
- SHA-256: `4ccbf9f1c22b5d21e6c6999d7285fade2fa7a845746ca9698a38431416c0bd44`  
- Source modified: `2026-06-19T16:50:19+00:00`  
- Imported at: `2026-07-05T16:48:25+00:00`  
- project: `wf_e27060ad-65f`  
- session_id: `8d4866bf-8edd-4da9-909e-022ddee12675`
```

Transcript

1. user (2026-06-19T16:44:38.374Z)

Assess PR #245 for convention compliance and strategic priority fit. Read these in the indexer repo: docs/PROJECT_DOC_TEMPLATE.md, docs/NEXT_STEPS.md, docs/README.md (doc index ? does RALLY_DETECTION_QUALITY_REPORT.md exist?), and skim docs/HOW_RALLY_DETECTION_WORKS.md / docs/MULTI_SIGNAL_FUSION_PLAN.md headers. Also note CLAUDE.md conventions (one PR per work item, no stacking; substantial design work gets a tracked project doc; Launch Report for default flips; decision-rigor norm).

The PR adds two docs: RALLY_DETECTION_CODE_REVIEW.md (a code review) and RALLY_DETECTION_FIXES_PLAN.md (a project-doc-template plan proposing 3 fixes: A _spread edge case, B _trim_inactive_tail density, C cvat parser logging). Body: "Once approved, I will implement the code changes."

TASKS ? fill every schema field:

1. template_compliance: Does the plan follow PROJECT_DOC_TEMPLATE.md? It contains sections 0,1,4,6,7,8,9,10 but SKIPS 2,3,5 ? identify what's missing (e.g. success metrics, alternatives, testing). Does it cite "RALLY_DETECTION_QUALITY_REPORT.md" as an existing doc ? does that file actually exist in the repo? (If not, that's a fabricated cross-reference.)
2. launch_report_applicability: Do any fixes change default/production behavior? Fix B changes boundary/density math ? would that shift rally boundaries and thus require a measured Launch Report per the owner's convention, not a blind code edit? Fix A/C are observability/edge-only ? classify each.
3. decision_rigor_alignment: Does the plan commit to measuring ?F1 / dual-set no-regression (per "NUMBERS INFORM, FOUNDATIONS DECIDE"), or does it just assert hardening with no measurement? Its DoD says "Tests updated and passing" ? is that sufficient for a boundary-behavior change?
4. roadmap_fit: Per NEXT_STEPS.md and the rally-quality track, what are the owner's actual top priorities (rally-END R4 end-rule, prominent-court, multi-signal fusion, Roora golden set)? Are these 3 micro-fixes aligned, orthogonal, or a distraction? Does fix B accidentally touch the #1 rally-END gap (and if so, should it be folded into that tracked effort rather than a standalone "harden edge case")?
5. funding_framing: The owner literally asked "whether we should merge and FUND the project being proposed." Right-size this: is it a fundable project or a ~3-function localized cleanup? Is there ANY funding/resourcing ask in the PR text? Be direct that the framing is oversized vs the content.
6. merge_recommendation + bottom_line.

Return structured output.

2. assistant (2026-06-19T16:44:41.813Z)

I'll assess PR #245. Let me start by reading the PR contents and the relevant convention docs.

3. user (2026-06-19T16:44:46.224Z)

```
{"additions":107,"baseRefName":"master","body":"Please review the proposed plan and code review
```

```
for the rally detection fixes. Once approved, I will implement the code changes.", "deletions": 0,
"files": [{"path": "docs/RALLY_DETECTION_CODE_REVIEW.md", "additions": 50, "deletions": 0, "changeType":
:"ADDED"}, {"path": "docs/RALLY_DETECTION_FIXES_PLAN.md", "additions": 57, "deletions": 0, "changeType":
:"ADDED"}], "headRefName": "proposal/rally-detection-fixes", "state": "OPEN", "title": "docs: Proposal
for Rally Detection Fixes"}
```

4. user (2026-06-19T16:44:46.245Z)

```
1      # Next steps (live) ? owned-model + multisport
2
3      **Updated:** 2026-06-13 (**rally-detection quality track**: fusion + experiment-harness
design docs merged,
4      owner-gated ? see the dedicated section below, the desktop-agent pick-up). Prior:
2026-06-10 (audit?M0?Gemini-battery session; later same day the **UI-alpha track opened**:
5      ADR-009 ratified ? KhelSutra Guru / khelsutra.guru, private `khelsutra` repo, local-
first delivery ? see
6      `docs/UI_ALPHA_EXECUTION_PLAN.md`; **PR-1 async jobs (#16) MERGED 2026-06-11, PR #87**;
**PR-2
7      upload/download MERGED 2026-06-11, PR #99** ? live-E2E-verified (3-part chunked upload
MD5 byte-identity +
8      compile?browser-download); next: PR-3 identity. Same day: the **8?9 quality sweep
#90?#98 landed** ? ruff/mypy/
9      coverage-floor CI lanes, SQLite conn-close fix, config unknown-key warning, hybrid-twin
collapse into
10     `trajectory_hybrid.py`, WSL tilde+abspath live-path fixes; suite 388 green; LOVO 0.626
re-verified exact).
11     **Authoritative roadmap:**
12     `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` (+ ADR-008 for topology). **Latest blow-by-
blow:** memory
13     `current-state.md` / `session-handoff.md`. This file = the prioritized "what to do
next," refreshed each session.
14
15     ## ? Next-session / GPU-box pickup (2026-06-15)
16     **M0 in-container quick wins are COMPLETE:** ship-gate provisional blocker (**#170**),
train/eval split guard (**#171**), the **doc-consistency sweep** (**#174** ? corrected 16
drifted docs vs the code), and the **P3 person-feature fusion litmus machinery is already
present + suite-green** (`backend/eval/fusion_compare.py` + `ablation.py`: person-driven ?F1 /
bootstrap CI / paired sign test; the real-golden `skipif` litmus waits only on GPU-extracted
data). ? CI is red repo-wide due to a **GitHub Actions minutes/billing outage** (account-level,
not code) ? work is local-gated + `--admin`-merged until Actions is restored (Settings ? Billing
? Actions minutes / spending limit).
17
18     **Immediate next step ? the real P3 LOVO measurement** (this box has a GPU +
`torch+cu124` + `cv2`; videos and trajectory CSVs are present, but the GPU-extracted golden
features + `.rallies.csv` labels are not yet here):
19     1. **[owner / GPU box]** `python -m backend.eval.fusion_golden --manifest
output/human_lovo_manifest.json --out-dir "%USERPROFILE%\OneDrive\rally-annotator\golden-
data\features\2026-06-15-n6" ? produces `*_golden_features.json` for the 6 golden videos. (GPU
+ WSL + videos + trajectory CSVs + `.rallies.csv` labels are owner-side only.)
20     2. Artifacts land in the **shared cloud folder** `C:\Users\avidu\OneDrive\rally-
annotator\golden-data\` (OneDrive auto-syncs both ways). Convention + workflow:
**[`GOLDEN_DATA_SHARING.md`](GOLDEN_DATA_SHARING.md)**; tracked in **#175**.
21     3. **[CPU dev/agent session]** `python -m backend.eval.fusion_compare --features-dir
<shared>\features\2026-06-15-n6 --record` + `python -m backend.eval.ablation --features-dir
<shared>\... --granularity group` ? real person-feature ?F1; copy the JSONs into `output/` to
also pass the `skipif` litmus tests (`tests/test_ablation.py::test_litmus_reproduces_gate_a`).
22
23     **Also queued (owner-gated / pending input):**
24     - Ship-gate target calibration ? **#172** (owner decision; needs corpus growth).
25     - Tests reorganization ? blocked on the owner's grouping metric (provide it ? an agent
will do it).
26     - Restore CI ? the Actions outage is account-level (billing/minutes), not a repo/config
bug.
27
28     **Leaving (bulky / M2 ? no start without owner go):** ActionFormer (M0.4), event-index
```

DB migration (M0.1), Gemini-silver purge (M0.3), cost-to-Gen-2 benchmark + ByteTrack deprecation, multisport, repo rename, PMF execution.

29

30 **## Where we are (one paragraph)**

31 ADR-008 ratified ("repo split driven by productionization, not isolation"): training lives in-repo (`training/`)

32 behind a CI-enforced import wall; the featurizer is single-sourced (`backend/features/rally\_seq.py`, train==serve);

33 the ****Gen-0 harness ships**** (`python -m training.gen0.harness --manifest output/human\_lovo\_manifest.json --floor`)

34 eval_baselines/heuristic_lovo_n6.json --version <semver>) and produced artifact v0.1.0 (LOVO ****0.626****, floor passed,

35 sign test 5/6 wins p=0.109 ? honestly not significant, ship=False). The ****n=6** owned corpus is in the collector**

36 (262 rallies, Proprietary, commercial export = owned data only after the ShuttleSet reclassification). The ****Gemini**

37 battery is done** (see table) ? the moat is quantified.

38

39 **## The quality table that now drives strategy (IoU 0.5 vs human golden labels)**

40 | Path | Mahadevpura (clean-ish) | Kushagra (multi-court) |

41 |---|---|---|

42 | Heuristic (velocity windowing) | 0.657 | 0.157 |

43 | ****Gen-0 owned head**** (LOVO, n=6) | 0.744 | 0.561 |

44 | Two-stage WASB+Gemini refine | 0.83 | ? |

45 | ****Gemini wrapper**** (`gemini-flash-latest`) | ****0.93**** | ****0.66**** |

46

47 ****Reading:**** clean footage is commoditized (serve it with Gemini for cents). ****Multi-court** drops the commodity

48 wrapper to 0.66 ? that is the owned model's ship target** (its FPs are background-game pickups + dead-time merges,

49 the exact contrast-metric confound). Gen-0 is 0.10 behind with only n=6 ? a corpus-growth-sized gap, not an

50 architecture-sized one. Two-stage refine = the cost-efficient serving path for LONG tapes (uploads candidate clips

51 only); wrapper 503-flakiness (observed all session) makes the provider-fallback registry load-bearing.

52 Baselines tracked in `eval\_baselines/` (heuristic_lovo_n6.json, gemini_wrapper_2026-06-10.json).

53

54 **## Rally-detection quality track (design merged 2026-06-13, owner-gated; implementation in progress)**

55 ****This is the track a desktop/local agent is slated to pick up.**** Design docs landed (PR #112 + session). All implementation is ****owner-gated**** (per explicit in the plan); each step = ONE PR off `master`.

56

57 ****Hardening loop (T1-T5 positive tests + audit + finale) COMPLETE (2026-06-14, PRs #156/#157/#160/#161/#163 + #165 + final records).**** See archived

58 `docs/archives/CODE\_AUDIT\_AND\_TEST\_HARDENING-COMPLETED-2026-06-14.md` (Living Log with every pre/post/review) and `docs/archives/HARDENING\_LOOP\_HANDOFF-COMPLETED-2026-06-14.md` (full rules + STATE). Top-level now pointer stubs per #165 review. All 5 items + verification + archive done; clean slate.

59 - ****Docs:**** `docs/MULTI\_SIGNAL\_FUSION\_PLAN.md` (shuttle + person + audio, one fusion layer) .

60 `docs/EXPERIMENT\_HARNESS.md` (A/B + shadow + promotion gate, zero-regression) .

61 `docs/ROORA\_LABELING\_GUIDELINES.md` (v8) . `docs/HOW\_RALLY\_DETECTION\_WORKS.md` (2-layer mental model).

62 - ****Key finding (no re-investigation needed):**** the owner's "held-shuttle counted as a rally" bug is ****NOT a missing capability**** ? `backend/pipeline/detectors/rally\_gate.py` already implements the net-crossing ****"Gate A"***** (`apply\_gate(..., min\_crossings=2)`) that kills service-feed/held-shuttle windows, but it is ****only called from eval drivers**** (and there was no `min\_crossings` knob in `IndexingConfig`). ****Wiring Gate A into the live segmenter (now via fusion_hybrid) was the single cheapest, highest-confidence quality win.**** (P2 FusionSegmenter specifics: registry `fusion\_hybrid`, ****default-OFF****, seeds from substrate, persists `net\_crossings` + optional person features, optional served Gate A/B via config knobs;

adversarially reviewed; suite green. See `tests/test\_served\_gate\_a\_regression.py` for the honest finding that on production windowing Gate A is neutral/negative in some cases, hence kept opt-in with knob for safety, and the leaderboard revert note.)

```
63 - **Current status (2026-06-14):** P1 (person cue extraction to reusable
`PersonDetector`) + P2 (FusionSegmenter + Gate A signal + served handoff gating) + experiment
harness P1 DONE (adversarial review + NO-REGRESSION; suite green). **P3 next (this PR + follow-
ups):** learned fusion (person features ? harness `compare` LOVO lift on the 6 golden videos;
eager-person-compute optimisation); then P4 audio via hit_detector.
64 - **Pick-up order (each = ONE PR off `master`, owner approval first, flag-gated default-
OFF, promote only on measured LOVO):**
65 1. (P2 first-step, largely done via fusion_hybrid) Wire Gate A into production served
path + config knob (min_crossings); apply in hybrids at runtime. (Now available opt-in; served
neutral on production windowing per litmus ? kept opt-in with knob for safety.)
66 2. P3 first step (this work item): harness `compare` LOVO litmus for person-on variant
(players_in_court + two_sided + colocation etc. from `fusion_features`) on the 6 golden; add the
eager compute optimisation note/landing. Pure + harness (testable here).
67 3. P3/P4 continuation: audio cue integration + full learned promotion if lift proven.
68 - **Discipline (from EXPERIMENT_HARNESS.md):** every new cue **flag-gated, default
OFF**; promote to default **only on measured LOVO lift** through `run_regression.py` (exit
0/1/3); pinned `CONTROL` ? rollback = config flip. Harness re-uses ~80% existing code.
69 - ? Gate A, person cue logic, harness, and pure fusion paths are testable here; the
heavier real-video / golden LOVO / ablation confirmation waits on the GPU-extracted golden
features + `.rallies.csv` labels, which are not yet present in this checkout.
70 - **External due-diligence review inputs (2026-06-17) ? PICK UP AFTER current ideas are
exhausted; full details + citations in [`RALLY_QUALITY_RESEARCH.md`](RALLY_QUALITY_RESEARCH.md)
$5.6.** An owner-commissioned 31-reference review of the quality report *validated* the
methodology and surfaced these **future** levers (research inputs, not active work, not new
claims; verify citations before any submission):
71 1. **"First-mile" paper framing (highest-value):** stroke-classifiers (BST/DSTA) + the
big datasets (ShuttleSet ~36k strokes, Fine-Badminton ~27k actions) all **assume a pre-trimmed
rally clip already exists** ? we solve the ignored prerequisite. Pitch the paper as **the
missing untrimmed-video preprocessing pipeline that lets BST/MonoTrack run on raw footage.**
72 2. **Ground R2 in Action-Spotting / Precise-Event-Spotting**
(SoccerNet/TenniSet/T-DEED) ? turns R2 into a defensible publishable contribution.
73 3. **Add domain baselines for any submission:** MonoTrack + BST on ShuttleSet / Fine-
Badminton (we have only heuristic + Gemini today).
74 4. **Prefer TemporalMaxer over ActionFormer** for the M0.4 scorer swap (parameter-free
max-pool, ~2.8x fewer GMACs, small-N robust) ? see "M0.4 next increment" below.
75 5. **Court polygons ? pose ? the rally-END lever:** masking on-court players can
resolve *occluded end-of-rally states* (the #1 remaining gap vs Gemini), not just spectator
noise.
76 6. **Name the eval?serve lesson "pipeline distribution skew"** ? a citable paper
"lesson."
77
78 **Discipline reminder:** owner approval before starting any owner-gated item; one PR per
step; babysit the PR (poll/fix/reply) until merge observed, *then* advance.
79
80 ## Prioritized next steps
81 1. **[owner, in progress] GROW THE GOLDEN CORPUS** ? the single highest-leverage move;
every lever below feeds on it.
82 Priority order for new videos: **(a) multi-court badminton** (the target distribution
AND where the ship target
83 lives), **(b) ?3 matches per new sport ? table tennis first** (WASB transfers at F1
0.909; pickleball labels are
84 corpus-gold but not trainable until a clean MIT/Apache ball detector exists), (c)
capture variety per
85 `VIDEO_RECORDING_GUIDELINES.md`; **adult sessions only for the training pool**
(consent guardrail). Per video:
86 label ? `curate import-local --normalize --license Proprietary --fps <true fps>` ?
re-run the Gen-0 harness.
87 The paired sign test needs n: at n=10 with 9 wins, p?0.011 ? significance is
reachable this month.
88 2. **[owner decision] Set the ship-gate target** ? now informed: floor = heuristic 0.480
(necessary), **multi-court
89 reference = wrapper 0.66** (the number that makes the owned model worth serving),
```

```

clean-footage ceiling = 0.93
90      (not the target ? Gemini serves that tier). Suggested shape: "LOVO F1@tIoU0.5 ? 0.70
on multi-court folds with
91      paired-sign p<0.05 vs heuristic" ? owner to ratify/adjust.
92      3. **M0.4 next increment ($0/local):** swap the Gen-0 LogReg for a lightweight temporal
head (MIT, minutes on the
93      2070S ? NOT the heavyweight Gen-2 stack) inside the existing harness; per-video
threshold calibration (the
94      LOVO-vs-oracle gap is visible per fold in the harness output). **Prefer
TemporalMaxer** (parameter-free
95      max-pool, ~2.8x fewer GMACs, small-N robust) over ActionFormer/ASFormer per the
2026-06-17 external review
96      ('RALLY_QUALITY_RESEARCH.md` $5.6) ? better fit for the cheap/local/fast mandate.
97      4. **Multisport thread:** ingest Roora pickleball/TT CSVs ('import-local --normalize`);
merged-prototype LOVO across
98      badminton+TT once TT has ?3 matches; identify a clean pickleball ball detector. ? TT-
on-WASB serving stays gated
99      by C3 (provenance multiplies per sport).
100     5. **PMF validation (cheaper than any engineering month):** C13 academy interviews (+
the two added questions:
101     "multi-court footage nobody watches?" / "pay per-match to index it?"); camera pilot
at the warmest 2-3 academies
102     (consent at capture; demo served via the Gemini path or human-polish ? the reel
pattern exists:
103     `output/MahadevpuraSingles_highlights.mp4`). **Field kit COMPLETE (2026-06-12, owner
ratified the
104     physical-visit approach; reframed product-feedback-first per owner spec):**
105     `docs/PRODUCT_FIELD_ASSOCIATE_FLYER.md` (the recruiting flyer ? owner fills ?/contact
placeholders and recruits) +
106     `docs/FIELD_VISIT_PLAYBOOK.md` (the hired associate's visit playbook: coach + student
script, demo-after-Q6
107     rule, ground rules, Phase-2 record-and-show-back stretch goal) +
`docs/FIELD_VISIT_ANSWER_SHEET.md` (per-visit
108     capture form incl. demo-reaction + student blocks) + `docs/PMF_FIELD_SIGNALS.md`
(G/A/R anchors + **pre-registered
109     decision rules** ? PROCEED / PIVOT A-C / FALSIFIED thresholds fixed before the first
visit).
110     6. **Housekeeping:** ~~rotate the Gemini API key~~ **DONE 2026-06-10** (owner-verified
working; confirm old-key
111     deletion in AI Studio); **rename repo off "badminton"** (owner action,
112     in BACKLOG); collector=SoR confirmation; off-peak niceties: none pending ? the
battery completed.
113
114     ## Pointers / artifacts
115     - Roadmap: `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` . Topology: ADR-008
(`docs/archives/decisions/DECISIONS.md`) .
116     Study: `docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md` . Quality history:
117     `docs/archives/quality-iterations/2026-06-multivideo-lovo/`.
118     - Harness: `training/gen0/harness.py` (one command ? train ? LOVO ? gate ? artifact
bundle; weights gitignored,
119     manifest reviewable). Featurizer: `backend/features/rally_seq.py` (FEATURE_SCHEMA v1).
120     - Corpus: collector DB (system-of-record, 6 matches/262 rallies) . eval manifest
`output/human_lovo_manifest.json`
121     (gitignored) . durable trajectories + Roora files: `C:\Users\avidu\Projects\Annotation
Setup\Collect\Trajectories\` .
122     labels: `...\Collect\Golden Labelled\`.
123     - Gemini ops learnings: serial uploads (`indexing.ai_max_workers=1`) + chunk re-encode
124     (`indexing.ai_chunk_scale_width=1280`) + 6-attempt exponential generate retry ? a
cold/prepaid project burst-throttles
125     with a MISLEADING "credits depleted" 429; ramp gradually.
126     - ? No owned-model implementation beyond greenlit M0.x until owner greenlights the next
milestone. Committed next
127     PLATFORM slice = async job model (single-tenant).
128

```

5. user (2026-06-19T16:44:46.246Z)

```
1      <!--
2      PROJECT DOC TEMPLATE ? copy this file to docs/<YOUR_PLAN>.md and fill it in.
3
4      A "tracked project doc" is how this repo carries substantial design/project work: it can
be
5      ITERATED on, its progress TRACKED, and it is MARKED DONE against explicit deliverables,
then
6      archived. Standardizes the emergent pattern in docs/RALLY_QUALITY_RESEARCH.md §7 and
7      docs/GOLDEN_REGRESSION_FIXTURES.md.
8
9      Rules of thumb:
10     - §7 (the progress tracker) is the SOURCE OF TRUTH ? one row per independently-shippable
PR.
11     - Keep it HONEST: tag each claim [verified] (checked against code), [researched] (needs
sign-off),
12     or [design] (proposed). The doc reflects real shipped state, never aspirational.
13     - It POINTS to detail (code anchors, research artifacts), it does not duplicate it.
14     - Sections marked OPTIONAL are included only when relevant (e.g. Foundation/Threat-model
for
15     research- or security/privacy-heavy work). Delete sections that do not apply.
16     - Move to docs/archives/ once Status = DONE (per the archive policy: only terminal-state
work).
17     -->
18
19     # <Project / Plan Title>
20
21     > **Status:** `DRAFT` ? `<owner-gated? | in progress>` . **Owner:** `<name>` .
**Created:** `<YYYY-MM-DD>` . **Last updated:** `<YYYY-MM-DD>`
22     > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE)
23     > **Tracking anchors:** §7 progress tracker is the source of truth; indexed in
`docs/README.md`; pointer in memory `current-state` (and `docs/NEXT_STEPS.md` once work starts).
24     > **Relation to existing docs:** `<extends | supersedes | peer-of ?>`
25     > **Honesty note:** mark each claim `[verified]` / `[researched]` / `[design]`. Not
legal/financial advice where applicable.
26
27     ---
28
29     ## 0. TL;DR
30     <2?5 sentences: the goal, the approach, and the single most important caveat. Call out
any finding that is already actionable on its own.>
31
32     ## 1. Problem & goal
33     <Why now; the concrete user-facing outcome; what "good" looks like. Link the code/docs a
reader must read to get current.>
34
35     ## 2. Decisions locked
36     <Table ? capture answered questions so they are not relitigated.>
37
38     | # | Decision | Source / date | Implication |
39     |---|-----|-----|-----|
40     | D1 | ? | ? | ? |
41
42     ## 3. Foundation ? research / prior art *(OPTIONAL)*
43     <Include when the plan rests on external facts: cited research, legal/compliance,
benchmarks, prior-art survey. Mark researched-vs-verified; list any sign-off gates this
creates.>
44
45     ## 4. Design / architecture
46     <The actual plan. Data contracts, diagrams, and a REUSE MAP ? name the existing
modules/code this builds on so it is clear what is new vs reused.>
47
48     ## 5. Threat model / risk table *(OPTIONAL ? security / privacy / compliance work)*
```

```

49     <Who/what is protected vs NOT; residual risks stated plainly.>
50
51     ## 6. Honest limits ? what this does NOT do
52     <The false-confidence guardrail: what a green suite / a DONE status does NOT prove.>
53
54     ## 7. Deliverables & progress tracker    ? **source of truth**
55     Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. **One small PR per row**,
56     branched from `origin/master`, no stacking. Update the row (status + PR link) as work lands.
57
58     | ID | Deliverable | Depends on | Gated? | Status | PR |
59     |----|-----|-----|-----|-----|----|
60     | P0 | ? | ? | No | ? | ? |
61
62     ## 8. Open questions ? owner / external *(split blocking-gates from nice-to-knows; name
63     who decides)*
64     1. ?
65
66     ## 9. Definition of done
67     <Explicit acceptance checklist. When all ticked ? flip `Status ? DONE` and archive.>
68     - [ ] ?
69
70     ## 10. References
71     **Internal code anchors `[verified]`:** ``. **Internal docs:** ``.
72     **External:** ``.
73
74     ---
75
76     ### Changelog
77     - `` ? ``.
78
79

```

6. assistant (2026-06-19T16:44:49.294Z)

Now let me check the doc index and read the two PR-added docs plus the rally-detection docs.

7. user (2026-06-19T16:44:50.251Z)

```

1     - [ROORA_LABELING_GUIDELINES.md](ROORA_LABELING_GUIDELINES.md) ? Latest (v8) vendor
2     labeling guidelines mirrored from the collector's `docs/annotation/`, with the fusion-driven
3     court-calibration / singles-doubles additions marked `? ADDED (fusion)`.
4
5     - [RALLY_QUALITY_RESEARCH.md](RALLY_QUALITY_RESEARCH.md) ? *** Foundation doc** for
6     closing the local-vs-Gemini rally-detection gap: the measured gap + the code-grounded over-merge
7     diagnosis ("shuttle moving ? rally"), an inventory of the eval/experiment harness + its gaps,
8     the strengthened evaluation framework (segmental Edit/F1@k that make over-segmentation first-
9     class, golden-corpus + active learning, Gemini-agreement as a *shadow* signal), an
10    **adversarially-verified, cited external-research synthesis** (TAL/TAS boundary heads, racket-
11    sports rally-state cues, small-data, eval rigor), a **prioritized cheap?durable roadmap with
12    falsifiable success criteria**, and (§7) a **tracked project plan** ? work-item checklist,
13    sequencing, the single quality **number** attributed to the effort, and a **definition of done**
14    (project completes when all tiers land + the number is recorded). The plan that the
15    [QUALITY_ITERATIONS.md](QUALITY_ITERATIONS.md) log executes against; ties together
16    EXPERIMENT_HARNESS / MULTI_SIGNAL_FUSION_PLAN / ANALYZERS_AND_RECONCILERS /
17    OWNED_MODEL_IMPLEMENTATION_PLAN.
18
19    - [RALLY_DETECTION_GAP_CLOSURE.md](RALLY_DETECTION_GAP_CLOSURE.md) ? **Tracked execution
20    doc (2026-06-18)** for closing the remaining rally-detection accuracy gap **per video, before
21    launch**, driven by the growing golden corpus. First ground-truth clip (GX010141) reframed it:
22    boundaries are tight (R2/R3/R4 holds) ? the gap is **detection**: (G1) recall (local & Gemini
23    both miss ~half on hard clips, and *different* rallies ? complementary), (G2) precision (local
24    over-fires), (G3) **rally-END over-extension on the quick-return-after-point** (shuttle still
25    moving ? needs a *player-state* signal). Levers (owner-gated): player-kinematics + shuttle-in-
26    hand / sustained-invisibility end-rule with reverse-timeline backtrack, `rally_gate` Gate A
27    wiring, complementary fusion. **Peer** of `RALLY_QUALITY_RESEARCH.md` (execution tracker, not
28    research); §7 is the source of truth.
29
30    - [PROMINENT_COURT_DETECTION.md](PROMINENT_COURT_DETECTION.md) ? **Design + experiment
31    (2026-06-19, default-OFF, owner-gated flip)** for the **multicourt G2 over-fire**: identify the

```

****prominent (foreground) court**** (the one covering the majority of the frame) and keep only rallies whose shuttle lives inside it ? re-defining the shuttle signal `(x,y,visible)` ? `(x,y,visible-AND-inside-prominent-court)` ***before*** windowing, so a background/adjacent-court shuttle never forms a rally. 2D floor polygon (MVP) ? ****pseudo-3D play-volume**** (refinement, so high clears aren't clipped). Born from the GX010142 rally-#1 background-court FP; reuses `fusion\_features.point\_in\_polygon` + `FusionConfig.court\_polygon`. §8 progress tracker is the source of truth.

5 - RALLY_DETECTION_QUALITY_REPORT.md ? ****Technical-paper checkpoint (2026-06-17)**** of rally-detection quality: the honest measured state (over-segmentation milestone tolerance-F1 0.09170.323 at n=11, p=0.002; merge-rate 0.69470.150; first real-footage ground-truth at-par with START at Gemini parity), the task-faithful R2 metric, a detailed Gemini comparison, next steps + expected wins, areas yet to be explored, and a publishability / path-to-submission verdict. Renders to a polished PDF; the externally-shareable "toward a paper" summary of `RALLY\_QUALITY\_RESEARCH.md` + `QUALITY\_ITERATIONS.md`.

6 - EVALUATION.md ? How we measure and improve quality.

7 - EXPERIMENT_HARNESS.md ? Design (P1 SHIPPED) for A/B + shadow + SxS experimentation on rally-detection variants with ****zero production regression**** ? experiments are decoupled from deploy (new cues default OFF, promote only on measured LOVO lift, rollback = flip a flag). Composes the existing `backend/eval/` machinery (`calibration`, `regression`, `metrics`); implemented in `backend/eval/experiment.py`.

8 - QUALITY_ITERATIONS.md ? ****Living log**** of rally-detection quality iterations (hypothesis ? what shipped ? measured/expected effect ? lesson). The reverse-chronological record; terminal-state snapshots get archived under `archives/quality-iterations/`.

9 - ANALYZERS_AND_RECONCILERS.md ? Design (owner-gated, NOT started): a ****signal-fusion framework**** for rally detection ? producers emit confidence-scored ****signals**** at three temporal ****scopes**** (frame detectors / window analyzers / session analyzers), a learned ****scorer**** weights them, and an auditable ****report-first reconciliation**** pass lints the decision against the evidence. The spine: ****no special-casing in the decision path**** (signals?scorer, never `if/else`). Worked examples: a service-fault analyzer, a warm-up/practice span detector, and consuming a per-frame shuttle-state detector. A peer to `EXPERIMENT\_HARNESS.md` and `MULTI\_SIGNAL\_FUSION\_PLAN.md` (ADR-007 modular rally layer).

10 - PERSONALIZATION_PLAN.md ? Design (owner-adopted ****hybrid****; Phase-0 landing) for the ****per-user personalization feature on a generalization-first baseline****: a thin per-user ***tuning*** layer gated by the honest small-N statistics (a ****min-N promotion floor**** + a ****structural-guard exemption****, never a service gate) + a ****contribution flywheel**** (free-tier videos pool into the owner-trained baseline; ****a tool that gets better the more you help it****). Records the ****locked owner decisions**** (identity, storage, `min\_n` semantics, Gate-A default, free-tier pooling). Built so far: per-user promotion gate (#141) + held-out-USER learning curve (#142).

11

12 - GOLDEN_REGRESSION_FIXTURES.md ? ****Design (owner-gated, NOT started):**** video-free, non-attributable regression fixtures so a clean clone runs the rally-seg suite with ****no video/GPU/DB****. Two layers ? the post-detector freeze + ****dual gates**** (characterization + quality-floor) reusing the `backend/eval/` stack, and the ****SEALED-FIXTURES**** privacy filter (de-attribution-at-source, tiered public/key-gated, shuttle-only public tier). Carries a cited ****IN/US/EU legal memo**** (derived-data IP/privacy), a threat model, and a ****§7 progress tracker + definition-of-done****. Extends GOLDEN_DATA_SHARING.md.

13

14 ## Product / platform plans

15 - DESKTOP_APP_PLAN.md ? ****Tracked scoping doc (2026-06-18, owner-gated, [design] ? NOTHING built):**** package the local pipeline as a ****cross-platform desktop app**** for non-technical players ? plug in the camera's USB/SD card, click once, get back ****only the highlight reels****, fully on-device (no upload, no original-video bloat). The centerpiece is the ****de-WSL native-compute port**** (run WASB natively per-OS: Linux/Windows CUDA · macOS MPS/CPU ? drop WSL2), and its ****make-or-break unknown****: is CPU acceptable on a typical ***no-GPU*** enthusiast laptop (WASB is ~3.4× real-time on a laptop 2070S)? §7 tracker = ****P0 compute-feasibility spike ? P1 native detector ? P2 app shell ? P3 packaging/installers****. Realizes the ****L0 fully-local**** tier of VIDEO_LOCALITY_MODEL.md; is the ****LocalDesktop` ComputeBackend**** of PLATFORM_ARCHITECTURE.md §3.2; the smaller/export-clean owned model (OWNED_MODEL_IMPLEMENTATION_PLAN.md) is one answer to the compute crux.

16


```

17     ## Practical / ongoing
18     - [TRACKNET_WSL_SETUP.md](TRACKNET_WSL_SETUP.md)
19     - Golden set docs (`GOLDEN_SET`)
20     - Evaluation harness and calibration drivers (in `backend/eval/`)
21     - [GOLDEN_DATA_SHARING.md](GOLDEN_DATA_SHARING.md) ? How the GPU box and CPU dev/agent
sessions share golden artifacts (trajectory CSVs, `_golden_features.json`) via a OneDrive
folder; the GPU-extract ? CPU-litmus split. See ##175.
22
23     ## Conventions & templates
24     - [PROJECT_DOC_TEMPLATE.md](PROJECT_DOC_TEMPLATE.md) ? Reusable skeleton for a tracked
project doc: status header · decisions-locked · progress tracker (???) · definition-of-done
· archived on completion. Standardizes the emergent pattern in `RALLY_QUALITY_RESEARCH.md` §7
and `GOLDEN_REGRESSION_FIXTURES.md`. Use for any substantial design/project work.
25
26     ## Historical documents
27     Completed plans, old research, ADRs, and checkpoints live in [archives/](archives/).
28
29     - [CODE_AUDIT_AND_TEST_HARDENING.md](CODE_AUDIT_AND_TEST_HARDENING.md) ? COMPLETED
(2026-06-14; T1?T5 + full loop + verification per #162). Living record of the June 2026 plumbing
/ contract / architectural audit, 626-test infrastructure, positive test plans for the 5
identified issues, multi-layer regression protocol (including ablation/experiment layer), and
pre/post-change verification checklist. See `docs/archives/CODE_AUDIT_AND_TEST_HARDENING-
COMPLETED-2026-06-14.md` and `docs/archives/checkpoints/2026-06-14-hardening-loop-complete.md`
for the final archived state. (Top-level is now pointer stub per review; full content in
archives/.)
30
31     See `archives/README.md` for the umbrella policy and `archives/past_projects/README.md`
for classification of small tech debt / starter projects.

```

8. user (2026-06-19T16:44:51.311Z)

```

<!--
PROJECT DOC TEMPLATE ? copy this file to docs/<YOUR_PLAN>.md and fill it in.
-->

```

Refactor Rally Detection Edge Cases & Parser Stability

```

> Status: `IN PROGRESS` · Owner: `Agent` · Created: `2026-06-19` · Last updated:
`2026-06-19`
> Lifecycle: `DRAFT ? IN PROGRESS ? DONE ? archived`
> Tracking anchors: §7 progress tracker is the source of truth; indexed in `docs/README.md`.
> Relation to existing docs: Extends `RALLY_DETECTION_QUALITY_REPORT.md`.
---

```

0. TL;DR

The rally detection pipeline is largely robust, but a deep code review highlighted three minor edge cases/bugs: a potential divide-by-zero or zero-spread in `\_spread` for tiny sequences, a density calculation vulnerability for sparse tracks in `\_trim\_inactive\_tail`, and silent skipping of malformed rows in the CVAT parsing script. This plan outlines the fixes for these issues to increase stability.

1. Problem & goal

The current implementation of the candidate windowing handles normal inputs extremely well (including handling blobs), but could fail or behave erratically under specific conditions:

- Zero-spread calculation:** `\_spread` in `rally\_gate.py` evaluates to `0.0` for arrays of length 2, which could cause a zero deadband if used on short clips.
- Sparse trailing frames:** `\_trim\_inactive\_tail` calculates frame density based on *visible* frames. A long gap followed by a single active frame gives 100% density, artificially inflating the rally end.
- Silent failure on malformed CSV:** `load\_canonical\_csv` in `rally\_cvat.py` silently ignores rows without a valid `rally\_number`.

****Goal****: Fix these three issues to harden the rally detection and ingestion pipelines without changing the existing heuristic thresholds.

4. Design / architecture

The changes will be entirely localized to the identified functions:

- ``rally_gate.py``: Modify ``_spread`` to use ``math.ceil()`` or properly enforce a minimum subset size before trimming percentiles, ensuring it never inadvertently calculates a spread of 0 for non-identical values.
- ``tracknet_runner.py``: Update ``_trim_inactive_tail`` to factor the **temporal length** (or total expected frames in the trailing window) into the density calculation, rather than purely relying on ``win_count`` (the number of visible frames).
- ``rally_cvat.py``: Introduce ``logger.warning`` before the ``continue`` statement in ``load_canonical_csv`` to surface skipped rows during CVAT ingestion.

6. Honest limits ? what this does NOT do

This will NOT change the core AI validation (Phase 3) or the existing ``velocity_thresh`` values. It solely tightens up boundary calculation edge cases. It will not suddenly "fix" rallies that the tracking model missed entirely due to lighting or occlusion.

7. Deliverables & progress tracker ? ****source of truth****

Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. ****One small PR per row****, branched from ``origin/master``, no stacking. Update the row (status + PR link) as work lands.

ID	Deliverable	Depends on	Gated?	Status	PR
P0	Fix <code>`_spread`</code> and <code>`_trim_inactive_tail`</code> edge cases	?	No	?	?
P1	Add logging to <code>`load_canonical_csv`</code> for silent skips	?	No	?	?

8. Open questions ? owner / external

1. Do we want ``_trim_inactive_tail`` to fall back to the existing behavior if there are very few frames, or strictly enforce a minimum temporal density? (Assuming strict temporal density is preferred for accuracy).

9. Definition of done

- [] ``_spread`` bug fixed and tested.
- [] ``_trim_inactive_tail`` density calculation corrected.
- [] Warning log added for malformed rows in ``rally_cvat.py``.
- [] Tests updated and passing.

10. References

****Internal code anchors `[verified]`****

- ``backend/pipeline/detectors/rally_gate.py``
- ``backend/pipeline/detectors/tracknet_runner.py``
- ``src/parsers/cvat/rally_cvat.py``

****Internal docs****

- ``docs/RALLY_DETECTION_CODE_REVIEW.md``

9. assistant (2026-06-19T16:44:56.981Z)

The README doc index lists ``RALLY_DETECTION_QUALITY_REPORT.md``. Let me verify it actually exists as a file, check the code review doc, and skim the two rally docs' headers. Also verify the code anchors the plan references.

10. user (2026-06-19T16:45:01.246Z)

```
=== Does RALLY_DETECTION_QUALITY_REPORT.md exist on master? ===
docs/RALLY_DETECTION_QUALITY_REPORT.md
--- on disk: ---
-rw-r--r-- 1 avidu 197609 32871 Jun 19 21:46 docs/RALLY_DETECTION_QUALITY_REPORT.md
=== Code anchor existence ===
backend/pipeline/detectors/rally_gate.py
backend/pipeline/detectors/tracknet_runner.py
```

```
=== search for rally_cvat / tracknet_runner anywhere ===
backend/pipeline/detectors/tracknet_runner.py
tests/test_tracknet_runner.py
```

11. user (2026-06-19T16:45:02.039Z)

Rally Detection Code Review

1. How Rally Detection Works

The project identifies rallies through a combination of local trajectory tracking (e.g., TrackNet or WASB) and a series of heuristic gating/trimming rules, culminating in an AI-based refinement (Phase 3). Here's a breakdown of the pipeline:

Phase 2: Candidate Windowing (The Heuristics)

- Velocity Thresholding:** The trajectory detector outputs `(Frame, Visibility, X, Y)`. For each visible frame, the normalized velocity (distance divided by frame width per second) is calculated. Frames with a velocity exceeding `velocity\_thresh` (e.g., 0.02) are considered "active" (meaning the shuttle is in flight).
- Merging / Grouping:** Active frames within a certain dead time (`merge\_gap`, usually 2.0s) are merged into candidate windows.
- Splitting Over-merged Windows (W1):** If a window exceeds `max\_window\_duration`, the code splits it at the largest internal gap (dead-time) that is at least `min\_split\_gap`. If the window lacks gaps (the shuttle never stops moving), a `window\_hard\_cap` setting (if enabled) forcibly chops it in half at the temporal midpoint.
- Inactive Tail Trimming (R4):** Even after a rally finishes, a player might pick up the shuttle and toss it, creating a slow-moving trajectory tail. To prevent the window from overextending, the algorithm trims the end back to the last segment of the window that had a dense cluster of "fast" frames (velocity > `rally\_thresh`).
- Blob Fallback (R5):** If a window still vastly exceeds the cap after the W1 split and R4 trim (meaning it's an unusually continuous blob of motion), it is chopped down to the maximum cap limit and marked with a `forced\_cut` flag. This signals that the clip needs "rally-state" cues (like net-crossings) rather than a simple duration cap.
- Rally Gate (Gate A):** In `rally\_gate.py`, a heuristic filters out "false" rallies (like a player tossing a shuttle back to their opponent). It estimates the net position (median play axis coordinate) and counts how many times the shuttle crosses the net. A valid rally must have `min\_crossings` (e.g., 2).

Phase 3: AI Handoff

Valid candidate windows are padded (e.g., by 2 seconds on each end) and sent to a Generative AI Provider (like Gemini) along with a sports-specific prompt. The AI confirms the semantic boundaries, ensuring the window correctly matches a true rally, and also attempts to count the shuttle exchanges.

2. Deep Code Review & Potential Issues Found

I performed a deep review of `tracknet\_runner.py`, `trajectory\_hybrid.py`, `rally\_gate.py`, and the parsing logic in `sports-data-collector`. Here are the findings and hidden issues:

A. Subtle Calculation Artifact in `\_spread` (`rally\_gate.py`)

Observation: The `\_spread` function calculates a robust spread (ignoring the bottom 5% and top 5% of coordinates to eliminate outliers):

```
`lo = s[max(0, int(0.05 * (len(s) - 1)))]`
`hi = s[min(len(s) - 1, int(0.95 * (len(s) - 1)))]`
```

Hidden Behavior: Because `int()` truncates rather than rounds, for very short arrays (e.g., < 20 items), the 95th percentile will simply be the second-to-last item, and for `len(s) == 2`, the spread evaluates to strictly `0.0`.

Impact: This doesn't crash the pipeline, because the play axis is correctly estimated over the *entire* video's thousands of trajectory points. However, if this utility function is reused

on small window slices in the future, it will quietly collapse the spread to zero.

B. Sensitivity to Sparse Tracking in `\_trim\_inactive\_tail` (`tracknet\_runner.py`)

****Observation:**** The inactivity trimmer identifies the end of a rally by looking for the last frame where the trailing `trail\_frames` window has a sufficient density of fast-moving frames (``fast_sum / win_count >= min_density``).

****Hidden Issue:**** `win\_count` represents the number of *visible, active frames* in that time boundary, rather than the total elapsed time. If the detector completely loses track of the shuttle for 1.8 seconds and only picks up 1 fast frame right at the end of the window, `win\_count` will be very low (e.g., 1 or 2). Consequently, ``fast_sum / win_count`` could easily evaluate to 1.0 (100% density).

****Impact:**** Sparse tracking points at the very end of a clip might prevent the tail trimmer from cutting off dead-time correctly, artificially inflating the end of the candidate window. This is somewhat mitigated by the `merge\_gap` breaking trajectories, but it remains a boundary logic quirk.

C. Graceful Degradation on Incomplete CSV Validation (`rally\_cvat.py` in `sports-data-collector`)

****Observation:**** The `parse\_cvat\_boxes` function searches zip files for an `.xml` ending in `annotations.xml` or, failing that, *any* `.xml`.

****Issue:**** In ``load_canonical_csv()``, ``safe_int()`` is used to catch malformed `rally\_number` fields. If a row is entirely malformed (but doesn't break the CSV parsing), it just ``continue`s` silently. While this provides stability, a severely malformed CSV might silently drop half its rallies without failing the import gate. Logging a warning for skipped rows would improve observability.

D. Safe Slicing in the W1/R5 Blob Splitters

The W1/R5 gap-splitters use a recursive worklist:

```
`split_i = TrackNetRunner._midpoint_split_index(h)`  
`work.append(h[:split_i]); work.append(h[split_i:])`
```

Because it explicitly enforces ``len(h) < 2`` before splitting, ``split_i`` will always be at least 1, meaning the code safely guarantees it will never enter an infinite loop of empty arrays. This is an excellent, defensively written chunk of code.

12. assistant (2026-06-19T16:45:06.059Z)

The `RALLY\_DETECTION\_QUALITY\_REPORT.md` cross-reference is valid (the file exists). The `rally\_cvat.py` parser lives in the sibling `sports-data-collector` repo, not this one (fix C). Let me quickly verify fix B's claim against the actual code in `tracknet\_runner.py` to assess whether it's a production-behavior change, and skim the two rally doc headers.

13. user (2026-06-19T16:45:08.754Z)

```
=== _trim_inactive_tail def ===  
294:         inactivity_min_density: float = 0.34,  
381:         groups = [TrackNetRunner._trim_inactive_tail(g, tw, inactivity_min_density,  
402:     def _trim_inactive_tail(group: List[Tuple[int, float]], trail_frames: int,  
403:         min_density: float, rally_thresh: float) -> List[Tuple[int,  
float]]:  
408:         ``trail_frames`` window holds >= ``min_density`` fraction of fast frames; trim back  
to the  
413:         if trail_frames <= 0 or len(group) < 2:  
418:         fast_sum = 0  
421:             fast_sum += fast[i]  
422:             while frames[lo] <= frames[i] - trail_frames:  
423:                 fast_sum -= fast[lo]  
425:             win_count = i - lo + 1  
426:             if win_count > 0 and fast_sum / win_count >= min_density:
```

14. user (2026-06-19T16:45:10.954Z)

```
src/parsers/cvat/rally_cvat.py
```

```
---HOW header---
```

How rally detection works (the 2-layer mental model)

A short conceptual map of how we go from raw video to rally `[start, end]` segments ?
and, crucially, **which layer our annotation can improve.**

```
...
```

```
video ??? frames ??? [DETECTION] ??? shuttle (x,y,visible) per frame
                        ?
(WASB CNN,             ? per-frame trajectory
 frozen weights)       ?
                        ?
                        velocity signal (how fast the shuttle moves)
                        ?
                        [SEGMENTATION] ??? the 3 knobs we tune
                        ?
                        ?
rally windows [start, end] ??? IoU vs golden labels
```

```
...
```

Two layers ? keep them separate in your head

****1. DETECTION ? "where is the shuttle, this frame?"****

WASB is a CNN that, for each frame, outputs a heatmap ? the shuttle's `(x, y)` + a visibility/confidence. Its weights were learned (by gradient descent, on shuttle-coordinate labels) and we use them ****frozen****. We chose a **shuttle** detector on purpose: "shuttle in
===FUSION header===

Multi-signal rally fusion ? add a human (person) cue alongside the shuttle

****Status:**** ****P1 DONE**** (2026-06-13) ? the person cue is extracted to
`backend/pipeline/detectors/person\_detector.py` (`PersonDetector`), consumed by `yolo\_hybrid`
with no
behavior change. ****P2?P4 remain owner-gated.**** Phases below are sequenced but each is a separate
PR.
****Owner decisions captured**** in [§9](#9-open-decisions-for-the-owner).

```
> **Companion docs:** the 2-layer mental model is
[HOW_RALLY_DETECTION_WORKS.md](HOW_RALLY_DETECTION_WORKS.md);
> the modular-fusion architecture this builds on is **ADR-007**
([archives/decisions/DECISIONS.md](archives/decisions/DECISIONS.md));
> the permissive-dependency rule is **ADR-002 / ADR-005**; the audio cue (the #1 next cue,
already
> scaffolded) is [archives/research/AUDIO_RALLY_DETECTION_PLAN.md](archives/research/AUDIO_RALLY
_DETECTION_PLAN.md).
> The vendor labels that gate the learned path come from
[GOLDEN_SET_VENDORING.md](GOLDEN_SET_VENDORING.md)
> and the labeling spec in [R00RA_LABELING_GUIDELINES.md](R00RA_LABELING_GUIDELINES.md).
> **How each cue here is added/tested without production regression** ? A/B, shadow eval, and
the
> promote-only-on-lift discipline ? is [EXPERIMENT_HARNESS.md](EXPERIMENT_HARNESS.md) (every
P1?P4 cue
> below lands flag-gated, default OFF, promoted only on measured LOVO lift).
```

```
---
```

1. Context ? why this, why now

15. assistant (2026-06-19T16:45:22.591Z)

I have everything I need. Let me confirm the key facts:

- Fix B's `\_trim\_inactive\_tail` is in the **live production segmentation path** (called from the active grouping at line 381, with `inactivity\_min\_density=0.34` as a real default). Changing the density math from `fast\_sum / win\_count` (visible-frame count) to a temporal-length basis would shift where rally windows get trimmed ? i.e., it moves rally **end** boundaries. This is exactly the #1 rally-END gap the owner has tracked in `RALLY\_DETECTION\_GAP\_CLOSURE.md` (G3: rally-END over-extension) and the R4 end-rule lever.
- Fix A (`\_spread`) is, by the code review's own admission (finding A), latent/dead ? the function is only used over the whole video's thousands of points; the bug "doesn't crash the pipeline" and only bites "if reused on small window slices in the future." Pure observability/edge-only, zero current production effect.
- Fix C (`rally\_cvat.py`) lives in the **sibling `sports-data-collector` repo**, not this one ? so it cannot be implemented in this PR/repo, and it's pure logging (observability).
- The cross-reference to `RALLY\_DETECTION\_QUALITY\_REPORT.md` is **valid** (file exists). No fabrication there.

The plan skips template sections 2 (Decisions locked), 3 (Foundation - optional), and 5 (Threat model - optional). The notable non-optional omission is section 2; there's no success-metrics/measurement section, no alternatives, and the testing/DoD is just "Tests updated and passing" with no ?F1 / dual-set no-regression commitment.

16. user (2026-06-19T16:45:54.830Z)

Output does not match required schema: root: must have required property 'launch_report_applicability', root: must have required property 'decision_rigor_alignment', root: must have required property 'roadmap_fit', root: must have required property 'funding_framing', root: must have required property 'merge_recommendation', root: must have required property 'bottom_line'

17. assistant (2026-06-19T16:45:57.368Z)

I had a stray `<</merge\_recommendation>` closing tag inside the `template\_compliance` value that broke the parse. Let me resubmit cleanly.

18. user (2026-06-19T16:46:21.846Z)

Output does not match required schema: root: must have required property 'launch_report_applicability', root: must have required property 'decision_rigor_alignment', root: must have required property 'roadmap_fit', root: must have required property 'funding_framing', root: must have required property 'merge_recommendation', root: must have required property 'bottom_line'

19. assistant (2026-06-19T16:46:26.698Z)

There's still a stray `<</invoke>` closing tag at the end of `bottom\_line` (a leftover `<</invoke>` inside the value). The XML parser is treating the early stray tag as the close. Let me check ? actually the issue is a literal `<</invoke>` string ended up appended inside the bottom_line value in my previous call. Let me resubmit, being careful that no value contains any angle-bracket tag text.

20. user (2026-06-19T16:46:50.688Z)

Output does not match required schema: root: must have required property 'launch_report_applicability', root: must have required property 'decision_rigor_alignment', root: must have required property 'roadmap_fit', root: must have required property 'funding_framing', root: must have required property 'merge_recommendation', root: must have required property 'bottom_line'

21. assistant (2026-06-19T16:46:56.761Z)

The parser is still only seeing `template\_compliance`. The issue is likely a stray character in the `template\_compliance` value ? I used `sec 7` etc., but the real culprit is probably an embedded angle bracket. Let me re-examine: my `template\_compliance` value contains `RALLY\_DETECTION\_GAP\_CLOSURE.md sec 7` ? no brackets there. But wait, the schema error consistently reports only the FIRST property as present and all others missing. This pattern

means the harness is parsing the parameters but the first value is "swallowing" the rest, OR there's an unescaped special character.

Let me try writing the values to be maximally plain ASCII with no special punctuation that could be interpreted as markup, keeping each value short.

22. user (2026-06-19T16:47:18.753Z)

Output does not match required schema: root: must have required property 'launch_report_applicability', root: must have required property 'decision_rigor_alignment', root: must have required property 'roadmap_fit', root: must have required property 'funding_framing', root: must have required property 'merge_recommendation', root: must have required property 'bottom_line'

23. assistant (2026-06-19T16:47:27.348Z)

The harness keeps seeing only `template\_compliance`. This strongly suggests the parameters are being parsed but only the first one survives ? likely because the very long single-line `template\_compliance` value is hitting some length truncation, cutting off the remaining parameters. Let me drastically shorten every field to test whether brevity fixes the schema validation.

24. user (2026-06-19T16:47:40.780Z)

Output does not match required schema: root: must have required property 'launch_report_applicability', root: must have required property 'decision_rigor_alignment', root: must have required property 'roadmap_fit', root: must have required property 'funding_framing', root: must have required property 'merge_recommendation', root: must have required property 'bottom_line'

25. assistant (2026-06-19T16:47:47.326Z)

Brevity didn't fix it, so it's not length. The harness is parsing only the first parameter. The likely culprit: my `template\_compliance` value contains a forward slash sequence `src/parsers/cvat/rally\_cvat.py` or some character that the tool-input parser mis-handles, OR the issue is the parameter delimiter itself.

Let me test the structural hypothesis directly: submit with the SIX named fields FIRST as trivial one-word placeholders and `template\_compliance` LAST. If the schema validates, the problem is positional/parsing of the first long field; if it still fails, the problem is elsewhere.

26. user (2026-06-19T16:47:48.628Z)

Structured output provided successfully

27. assistant (2026-06-19T16:47:52.307Z)

The order/structure works ? the schema accepted all seven fields when ordered this way. The earlier failures were a parsing artifact of the field ordering/long first field. Now let me resubmit with the full real content in this same field order.

28. user (2026-06-19T16:48:16.924Z)

Output does not match required schema: root: must have required property 'template_compliance', root: must have required property 'decision_rigor_alignment', root: must have required property 'roadmap_fit', root: must have required property 'funding_framing', root: must have required property 'merge_recommendation'

29. assistant (2026-06-19T16:48:20.264Z)

I had malformed tags in that call (the `` and `` text). The successful test proved the field order works. Let me resubmit

carefully with clean tags, in the same order that just succeeded: the six non-template fields first, template_compliance last.

30. user (2026-06-19T16:48:44.765Z)

Output does not match required schema: root: must have required property 'template_compliance', root: must have required property 'decision_rigor_alignment', root: must have required property 'roadmap_fit', root: must have required property 'funding_framing', root: must have required property 'merge_recommendation', root: must have required property 'bottom_line'

31. assistant (2026-06-19T16:48:50.157Z)

The first parameter (`<launch_report_applicability>`) closed correctly with `</launch_report_applicability>`, but subsequent parameters used `<parameter name="...">` ... `</parameter>` which is wrong ? each parameter must open AND close with its own field-name tag. In my successful test, every parameter used matching name tags. The mangling happened because I wrote `</parameter>` as the closing tag. Let me redo with each field opening and closing on its own name, exactly like the successful test.

32. user (2026-06-19T16:49:14.125Z)

Output does not match required schema: root: must have required property 'decision_rigor_alignment'

33. assistant (2026-06-19T16:49:17.973Z)

Progress ? only `<decision_rigor_alignment>` is missing now. The problem is that field's value ends with a literal `</parameter>` in my call (I wrote `<tests only.</parameter>` instead of closing with the field name). The closing tag must match the opening name. Let me fix just that one field's closing tag by resubmitting the whole thing with correct closing tags everywhere.

34. user (2026-06-19T16:49:42.180Z)

Output does not match required schema: root: must have required property 'template_compliance', root: must have required property 'decision_rigor_alignment', root: must have required property 'roadmap_fit', root: must have required property 'funding_framing', root: must have required property 'merge_recommendation', root: must have required property 'bottom_line'

35. assistant (2026-06-19T16:49:46.899Z)

I keep introducing mismatched closing tags (e.g. `</merge_recommendation>` opening was `<parameter name="merge_recommendation">`). The successful test used `<parameter name="X">...</parameter>` style consistently ? opening with `<name=` and the harness matched them. The mismatches break everything. Let me replicate the EXACT format from my successful test: every field opens with `<parameter name="FIELD">` and closes with `</parameter>`. I'll be meticulous this time.

36. user (2026-06-19T16:50:11.063Z)

Structured output provided successfully

37. assistant (2026-06-19T16:50:19.326Z)

Structured output submitted successfully. Assessment of PR #245 complete.

Summary of findings (all returned via the StructuredOutput tool):

- **Template compliance:** Partially compliant ? uses the template but omits the non-optional \$2 (Decisions locked) and has no success-metrics/measurement section or alternatives. The `RALLY_DETECTION_QUALITY_REPORT.md` cross-reference is *not* fabricated (file exists, indexed in README), but the "Extends" relation is mis-stated, and the Fix C anchor `src/parsers/cvat/rally_cvat.py` is mis-homed (it lives in the sibling `sports-data-collector` repo, not this one).

- **Launch Report:** Fix B (`\_trim\_inactive\_tail`) is live-path and moves rally-END boundaries ? requires a measured Launch Report. Fixes A and C are latent/observability-only ? no report needed.
- **Decision rigor:** Misaligned ? zero ?F1 / dual-set no-regression commitment; "tests passing" is insufficient for the boundary-behavior change in Fix B.
- **Roadmap fit:** Mostly a distraction; Fix B accidentally touches the #1 rally-END gap (G3) and should be folded into the tracked `RALLY\_DETECTION\_GAP\_CLOSURE.md` effort rather than shipped standalone.
- **Funding framing:** Badly oversized ? no funding ask exists in the PR; it's a ~3-spot localized cleanup, not a fundable project.
- **Recommendation:** merge-with-changes.